

A Generator–Realizer Architecture for Constrained Predictive Physical Human–Robot Interaction

Draft status. This document is a new research direction, not a revised presentation of the existing double-integrator MPC paper. The numerical claims in Sections 7 and 8 are generated by the code in this folder: Section 7 is a planar point-mass proof of concept, and Section 8 is the same generator–realizer architecture instantiated on a torque-controlled 7-DOF Franka FR3 in MuJoCo. The FR3 study is a focused core (impedance and admittance generators, horizon-wide torque feasibility, slack-relaxed workspace/speed bounds, a reactive-clipping comparator, one benchmark scenario) rather than the full validation program a manipulator-level claim would eventually need; human-participant, hardware, and comparisons against the broader predictive-impedance literature remain future work (Section 8.4).

Abstract

Physical human–robot interaction controllers typically couple two decisions that are conceptually separate: what interaction behavior is desired, and how a constrained robot delivers it. This paper separates them architecturally. A pluggable interaction-dynamics generator specifies a desired contact-port acceleration $a_k^{\text{id}} = f_\theta(e_k, \dot{e}_k, F_{h,k})$; a receding-horizon realization layer tracks that behavioral acceleration — not a motion trajectory — under actuator, command-rate, speed, and workspace constraints, and exposes the gap between what was requested and what the constrained robot delivered as an explicit realization residual. For the memoryless affine generator class, the decision variable, feasible set, and functional form of the quadratic program are fixed, while the generator changes the objective coefficients through (C_θ, G_θ) . We state this template-invariance property precisely and validate it with linear impedance and force-guided admittance.

In a reproducible planar simulation, the same predictive layer is used without retuning for both generators. When the unconstrained reference behavior would leave a 0.10 m workspace, instantaneous realization followed by command clipping reaches 0.266 m for the impedance generator and 0.525 m for the admittance generator. Predictive realization limits displacement to 0.100002 m and 0.0912 m, respectively, while respecting the 0.22 m/s component-wise speed bound to numerical tolerance. The required deviation from the requested behavioral acceleration is exposed as a measurable realization residual rather than hidden by parameter adaptation. A further experiment

reassigns the generator on one running controller instance — impedance, then admittance, then impedance — and verifies a rate-bounded transition without controller reconstruction.

We then instantiate the same architecture on a torque-controlled 7-DOF Franka FR3 in MuJoCo, freezing the task-space inertia, feedforward torque, and Jacobian at each solve (a local approximation, not exact horizon-wide prediction) and enforcing per-joint torque feasibility at every predicted step. Under a sustained 20 N push toward a 0.06 m/0.35 m/s component-wise workspace/speed bound, predictive realization holds displacement to within roughly 0.1--0.2 mm of the bound via a slack-relaxed soft constraint, while reactive clipping overshoots by 4.4 cm (impedance) and 4.7 cm (admittance). We report both the residual predicted by the local model and an empirical residual obtained from the MuJoCo end-effector velocity. These results establish a focused simulation proof of the architecture, not hardware safety or optimality of the separation.

Keywords: physical human–robot interaction, generator–realizer architecture, model predictive control, behavior-realization separation, impedance control, admittance control, constraint handling.

1. Introduction

A pHRI controller has two conceptually different responsibilities:

1. specify how the robot should move in response to contact; and
2. decide what motion can safely be executed by the physical robot.

Hypothesis. For the memoryless affine generator class of Definition 1 (Section 3.3), interaction behavior can be supplied through a common desired-acceleration interface while the robot-level decision variable and feasible set remain unchanged. Theorem 1 proves this structural property. It does not claim that fixing the generator is lossless relative to jointly optimizing behavior parameters and robot commands.

Conventional impedance control combines these responsibilities in one mass–spring–damper feedback law. Variable-impedance methods add adaptation, and model-predictive variants can optimize impedance parameters or jointly plan motion and compliance. This is useful, but the behavior representation and the constraint-handling controller remain tied to impedance parameters.

A concrete consequence of that coupling is actuator saturation. A conventional impedance or PD law computes joint torque directly from the interaction error, $\tau = J^\top(-K_d e - D_d \dot{e}) + \tau_{\text{ff}}$, where τ_{ff} is the feedforward gravity/Coriolis compensation and, on a redundant arm, the orientation-hold and null-space terms as well. Neither the impedance term nor τ_{ff} reasons about the actuator's torque limit. A stiff enough impedance, a large enough interaction force, or simply an unfavorable configuration in which τ_{ff} itself is large can drive the commanded torque past τ_{max} . When the actuator clips, the

acceleration the impedance law intended is silently not the acceleration the robot delivers. This paper instead includes the *total frozen-model torque* — feedforward, orientation, null-space, and interaction terms — as a constraint at every prediction step (Section 8.1). Saturation is therefore anticipated within the local model, and the required deviation from the generator is represented by the realization residual. Because the manipulator prediction is frozen at each solve, this is not a global nonlinear-plant guarantee; executed torque is also monitored at the 1 kHz inner loop.

We study a different decomposition:

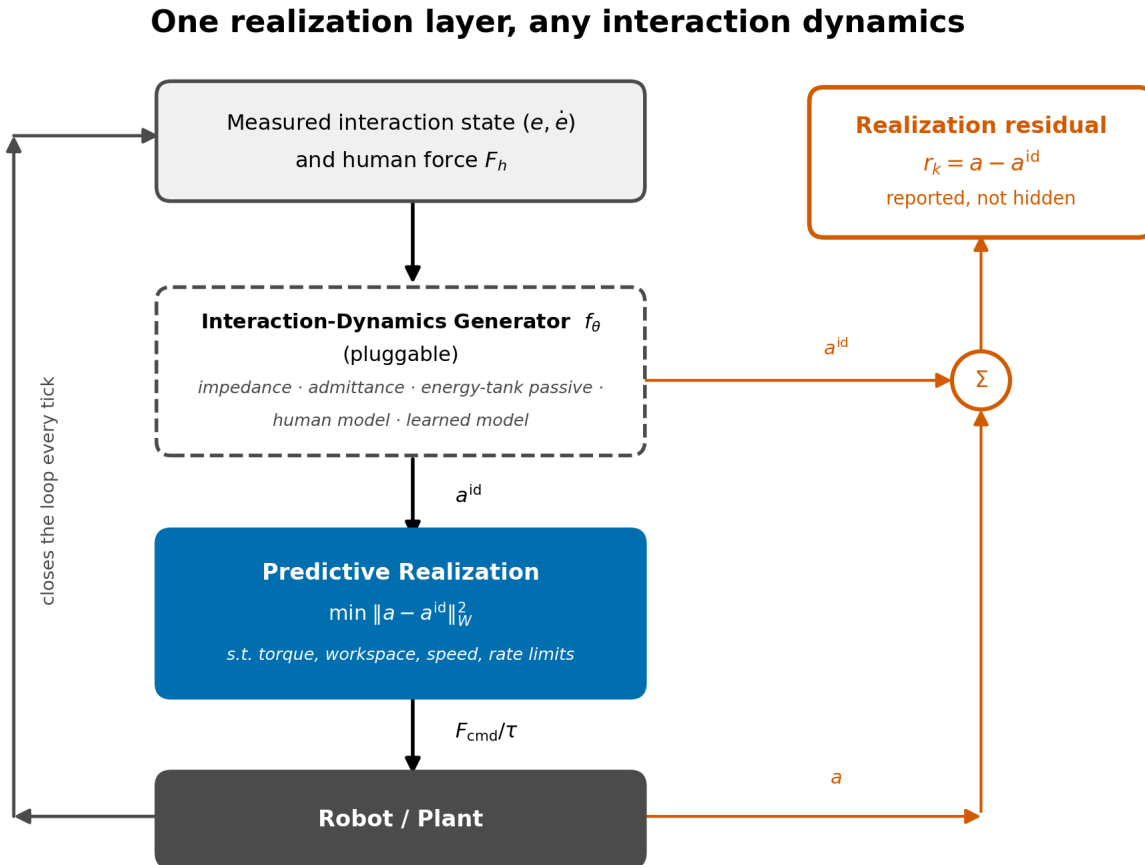


Figure 1. The generator (pluggable, dashed border) supplies a desired behavioral acceleration law a^{id} ; the realization layer (blue) retains one QP template and receives the generator-dependent law through its objective. The realization residual $r_k = a - a^{id}$ (right) is reported explicitly rather than absorbed silently into the command.

The generator expresses the intended interaction. The predictive layer is not asked to choose stiffness or damping. It chooses the actual robot command, and it is permitted to depart from the reference dynamics only when required by physical constraints or secondary objectives. The magnitude and timing of that departure are explicit in the *realization residual*.

This is what the hypothesis looks like as an architecture. For affine generators, the realization layer retains one decision variable, feasible set, and objective template; the requested behavioral

acceleration enters through (C_θ, G_θ) , which changes the numerical Hessian and gradient of the residual cost. Section 6 states this as the Affine Generator Template Invariance theorem. Section 4.1's impedance generator and Section 4.2's admittance generator are validated instances. Nonlinear, stateful, energy-tank, human-model, and learned generators retain the conceptual acceleration interface but may require linearization, state augmentation, additional constraints, or a nonlinear optimizer.

This separation changes both the algorithm and the scientific question. Instead of asking whether MPC can produce compliant tracking, we ask:

How accurately can a constrained robot realize a specified interaction dynamical system, and what controlled deviation is necessary when that system is temporarily incompatible with safety constraints?

The contributions of this draft are:

- a generator–realizer architecture whose robot decision variable, feasible set, and objective template are invariant across memoryless affine interaction generators, stated formally in Section 6;
- a convex predictive formulation for affine interaction dynamics, validated with impedance and force-guided admittance as two interchangeable instances of one generator interface;
- an explicit realization residual that quantifies how far the constrained robot had to depart from the specified dynamics, and why;
- a reproducible simulation, on both a planar point mass and a torque-controlled FR3 manipulator in MuJoCo, comparing horizon-wide constrained realization against instantaneous reference realization with identical command limits;
- an online generator-switching demonstration (Section 7.3) in which one controller instance accepts a different affine generator mid-run and the existing command-rate constraint bounds the transition.

We do **not** claim that embedding impedance behavior in MPC is itself new. Model predictive impedance control and model predictive interaction control already provide important prior formulations. The narrower contribution is an explicit desired-acceleration interface and realization residual together with template invariance for memoryless affine generators. Extension to nonlinear or stateful generators is future work, not covered by Theorem 1.

2. Related Work and Positioning

The closest literature is best distinguished by what the predictive or adaptive layer optimizes and by what quantity the robot-level controller is asked to reproduce.

Class	Behavior representation	Optimized robot-level variable	Role of prediction
Predictive variable impedance	M_d, D_d, K_d , often with a trajectory	Impedance parameters and/or trajectory	Select compliant behavior
Reference-model adaptive impedance	$M_d\ddot{e} + D_d\dot{e} + K_de = F_h$	Adaptive feedback parameters	Usually absent
Direct robot MPC	State or tracking-error dynamics	Force, torque, velocity, or acceleration	Optimize robot motion directly
Impedance/interaction MPC	Impedance, force, or coupled interaction model	Robot command and sometimes behavior parameters	Embed interaction objectives and constraints
Reference governor	A single reference/setpoint signal	Filtered reference fed to a fixed inner-loop controller	Supervise a non-predictive inner loop to enforce constraints
This work	$a^{\text{id}} = f_\theta(e, \dot{e}, F_h, z)$	Robot command u	Minimize dynamics-realization error under constraints

2.1 Predictive variable-impedance adaptation

Several approaches use MPC to adapt impedance parameters or to plan impedance and motion together. Anand et al. place a learned predictive policy above a low-level variable-impedance controller [1]. Haninger et al. predict interaction and optimize trajectory and impedance online while incorporating safety constraints [2]. Recent predictive variable-impedance formulations continue this

parameter-adaptation direction [3]. Their optimized behavior variables are stiffness, damping, or related trajectory parameters.

2.2 Reference-model impedance control

Model-reference adaptive impedance control specifies desired impedance dynamics and designs an adaptive nonlinear controller so that the robot approaches them [4]. This establishes the value of separating a desired model from the physical robot, but it does not by itself provide horizon-wide handling of coupled state and input constraints.

2.3 Direct robot MPC

A separate class applies MPC directly to the robot or to a feedback-linearized error model. Its decision variables are robot commands rather than impedance parameters. Such a controller can generate an effective closed-loop impedance, but unless a desired interaction model appears explicitly in the prediction objective, that impedance is an induced property of the optimizer rather than the behavior it is asked to realize. This distinction separates the present formulation from our earlier double-integrator tracking MPC: the earlier controller penalizes interaction error, whereas the present controller penalizes disagreement with an independently specified interaction dynamical system.

2.4 Model predictive impedance and interaction control

Bednarczyk et al. formulate impedance behavior through MPC and handle practical constraints such as velocity, energy, and jerk [5]. Gold et al. formulate model predictive interaction control using robot and interaction models in the optimal-control problem and express manipulation objectives through costs and constraints [6]. Minniti et al. combine whole-body MPC with online adaptation of robot–environment interaction models for constrained mobile manipulation [7]. These papers are close precedents and make a broad “first constrained interaction-dynamics MPC” claim untenable.

2.5 Reference governors and supervisory constraint enforcement

Reference governors are the closest architectural relative to this paper's separation, from a different literature than Sections 2.1–2.4. A reference governor sits upstream of a fixed, already-designed inner-loop controller and modifies the reference signal so that constraints on the resulting closed-loop trajectory are respected [8]. The parallel is direct: a reference governor also separates what is commanded from what is safe to execute. The difference is where prediction lives and what is exchanged. A reference governor typically supervises a fixed inner loop and manipulates a reference

signal; the realization layer here is itself predictive, and the exchanged object is an affine desired-acceleration law evaluated over predicted interaction states. The realization residual additionally measures the difference between requested and delivered dynamics. This connection limits the novelty claim: the contribution is not separation in the abstract, but its formulation and measurement at the interaction-dynamics level.

2.6 Position of this work

Every row above except the last two optimizes a quantity tied to one behavior representation: impedance parameters, adaptive impedance gains, or robot-level tracking against a specific interaction model. The reference-governor row is architecturally closest. Our narrower distinction is that a memoryless affine desired behavioral acceleration is exposed through a common interface, while the robot constraints are independent of the selected generator. This draft validates that interface for impedance and admittance. Demonstrating broader generality requires nonlinear/stateful generators and matched comparisons with reference governors, predictive impedance control, and model predictive interaction control.

3. Problem Formulation

3.1 General robot and contact-port dynamics

Consider a rigid manipulator

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = \tau + J(q)^\top F_h,$$

where $q \in \mathbb{R}^{n_q}$, $\tau \in \mathbb{R}^{n_q}$, and F_h is the human wrench applied at the interaction port. For a task coordinate $y = h(q) \in \mathbb{R}^{n_y}$,

$$\dot{y} = J(q)\dot{q}, \quad \ddot{y} = J(q)\ddot{q} + \dot{J}(q, \dot{q})\dot{q}.$$

Substitution of the joint dynamics gives the affine acceleration map

$$\ddot{y} = b_y(q, \dot{q}, F_h) + G_y(q)\tau,$$

where

$$\begin{aligned} b_y(q, \dot{q}, F_h) &= JM^{-1}(-C\dot{q} - g + J^\top F_h) + \dot{J}\dot{q}, \\ G_y(q) &= JM^{-1}. \end{aligned}$$

Let y_d be a nominal interaction pose and define

$$e = y - y_d, \quad \dot{e} = \dot{y} - \dot{y}_d.$$

Definition 1 (interaction-dynamics generator). A generator is a causal map

$$\ddot{e}^{\text{id}} = f_\theta(e, \dot{e}, F_h, z),$$

where z denotes optional internal generator state and θ denotes behavior parameters. Its output is a desired contact-port acceleration — the *behavioral acceleration* referred to throughout this paper — not a robot command. The generator is deliberately unaware of robot torque limits, joint limits, or workspace geometry; those belong to the realization layer.

The proposed architecture does not ask the generator to compute τ . Instead, it supplies $\ddot{e}^{\text{id}}$ above, and the predictive realization layer selects τ so that the physically realized $\ddot{e}$ approaches $\ddot{e}^{\text{id}}$ while satisfying robot constraints.

The full manipulator formulation is the target architecture. The executable study in this draft uses the following exactly discretized point-mass specialization so that the generator–realizer distinction can be isolated without kinematic or inverse-dynamics confounds.

3.2 Planar proof-of-concept plant

The proof-of-concept plant is a planar point mass:

$$m_r \ddot{p}(t) = u(t) + F_h(t),$$

where $p \in \mathbb{R}^2$ is displacement from a nominal interaction pose, $u \in \mathbb{R}^2$ is commanded robot force, $F_h \in \mathbb{R}^2$ is measured human force, and $m_r > 0$ is the realized robot mass. Define

$$x = \begin{bmatrix} p^\top & v^\top \end{bmatrix}^\top, \quad v = \dot{p}.$$

Under zero-order hold with period Δt ,

$$x_{k+1} = Ax_k + B(u_k + F_{h,k}),$$

$$A = \begin{bmatrix} I & \Delta t I \\ 0 & I \end{bmatrix}, \quad B = \begin{bmatrix} \frac{\Delta t^2}{2m_r} I \\ \frac{\Delta t}{m_r} I \end{bmatrix}.$$

The actual acceleration is

$$a_k = \frac{u_k + F_{h,k}}{m_r}.$$

3.3 The generator specialized to the planar plant

For the planar plant of Section 3.2, Definition 1 specializes with $x_k = [p_k^\top v_k^\top]^\top$ in place of $(e, \dot{e})$:

$$a_k^{\text{id}} = f_\theta(x_k, F_{h,k}, z_k).$$

For the affine class considered here,

$$a_k^{\text{id}} = C_\theta x_k + G_\theta F_{h,k}.$$

3.4 Realization residual

We define

$$r_k = a_k - a_k^{\text{id}}.$$

If $r_k = 0$, the robot exactly realizes the requested behavioral acceleration at that sample. If $r_k \neq 0$, the robot executes the closest feasible dynamics selected by the optimizer. Reporting r_k distinguishes unavoidable safety intervention from failure to tune the reference model.

This definition is intentionally stronger than commanding a nominal impedance force and penalizing

$$\left\| u_k - u_k^{\text{imp}} \right\|_2^2.$$

The latter is controller-output tracking: it penalizes deviation from a nominal force command. $\|r_k\|_W^2$ is tracking too, but of a categorically different target — the *behavioral acceleration* the generator specifies, not a prior controller's output — so it compares the realized physical acceleration directly with the requested behavioral acceleration. It remains meaningful when the generator is admittance-based, nonlinear, energy-based, or learned and no u_k^{imp} exists.

4. Configurable Generators

Section 3.1's Definition 1 specifies the generator interface abstractly; Section 3.3 specialized it to the planar plant. This section instantiates it.

Generator	What changes in the QP	Status
Impedance (4.1)	C_θ, G_θ only	Validated, Sections 7–8
Force-guided admittance (4.2)	C_θ, G_θ only	Validated, Sections 7–8

Generator	What changes in the QP	Status
Nonlinear/energy-tank, human-model, learned (4.3)	Not a (C_θ, G_θ) instance of this QP; would need generator-state augmentation, linearization, or a nonlinear optimizer	Sketched only, not implemented

The two validated rows use the same Section 5 affine QP template, differing only in (C_θ, G_θ) per Theorem 1. Section 4.3 sketches three further generators that share the desired-acceleration interface but not the affine structure; they are a scoped research roadmap, not implemented instances, and are deliberately not given equal billing with the validated rows above.

4.1 Impedance generator

The desired impedance is

$$M_d a^{\text{id}} + D_d v + K_d p = F_h.$$

For isotropic parameters,

$$a^{\text{id}} = -\frac{K_d}{M_d} p - \frac{D_d}{M_d} v + \frac{1}{M_d} F_h.$$

Thus

$$C_{\text{imp}} = \begin{bmatrix} -K_d M_d^{-1} I & -D_d M_d^{-1} I \end{bmatrix}, \quad G_{\text{imp}} = M_d^{-1} I.$$

Unlike variable-impedance MPC (Section 2.1), M_d, D_d, K_d are not decision variables in the present formulation. They describe the requested behavior.

4.2 Force-guided admittance generator

The second generator requests a force-dependent velocity:

$$T_a a^{\text{id}} + v = Y F_h,$$

or

$$a^{\text{id}} = -T_a^{-1} v + T_a^{-1} Y F_h.$$

It has no position-restoring term. After force release, velocity decays and the displaced position is retained. This provides a qualitatively different behavior through the same acceleration interface.

4.3 Future nonlinear generators

The three sketches merged into Section 4's catalog table are not variants of one equation; each replaces the affine law differently. A nonlinear passive dynamics, with a general (not necessarily linear) damping term and a potential function in place of a linear spring,

$$a^{\text{id}} = M_d^{-1} (F_h - D(v)v - \nabla V(p)),$$

is one instance. An energy-tank-modulated version adds a tank state that bounds the energy the generator may inject. A human internal/intent model or a learned interaction model instead replaces f_θ with an estimator or trained model. These extensions preserve the desired-acceleration interface but may require state augmentation, sequential linearization, nonlinear MPC, uncertainty propagation, and additional constraints. They are not covered by the affine QP theorem.

5. Predictive Realization

Section 4 supplies a^{id} . For a memoryless affine generator it is represented by C_θ, G_θ , producing one condensed-QP template whose numerical cost coefficients change with the generator while its decision variable and feasible set do not. At time k , let

$$U_k = \begin{bmatrix} u_{0|k}^\top & \cdots & u_{N-1|k}^\top \end{bmatrix}^\top.$$

The controller solves

$$\begin{aligned} \min_{U_k} \quad & \sum_{i=0}^{N-1} \|a_{i|k} - a_{i|k}^{\text{id}}\|_W^2 + \lambda_u \|u_{i|k}\|_2^2 + \lambda_{\Delta u} \|u_{i|k} - u_{i-1|k}\|_2^2 \\ \text{s.t.} \quad & x_{i+1|k} = Ax_{i|k} + B(u_{i|k} + \hat{F}_{h,i|k}), \\ & a_{i|k} = (u_{i|k} + \hat{F}_{h,i|k})/m_r, \\ & a_{i|k}^{\text{id}} = C_\theta x_{i|k} + G_\theta \hat{F}_{h,i|k}, \\ & |u_{i|k}|_\infty \leq u_{\max}, \\ & |u_{i|k} - u_{i-1|k}|_\infty \leq \dot{u}_{\max} \Delta t, \\ & |p_{i+1|k}|_\infty \leq p_{\max}, \\ & |v_{i+1|k}|_\infty \leq v_{\max}. \end{aligned}$$

Only $u_{0|k}^*$ is executed. $u_{-1|k}$ in the $i = 0$ rate penalty and rate constraint denotes the command actually applied at the previous solve, carried by the controller between calls (zero before the first solve), so both are well-defined from the very first call. The implementation uses the measured force

held constant across the horizon, $\hat{F}_{h,i|k} = F_{h,k}$, rather than oracle knowledge of the scripted force. A learned or physics-based force predictor can be substituted without changing the generator interface.

5.1 Tracking behavioral acceleration, not a motion trajectory

The objective $\|a_{i|k} - a_{i|k}^{\text{id}}\|_W^2$ is tracking — of a categorically different target than a position or velocity trajectory. No desired position trajectory is constructed; the optimizer instead compares two accelerations evaluated at the predicted interaction state: the acceleration the constrained robot will produce and the behavioral acceleration the interaction model requests. Position and velocity enter as generator states and as safety-constrained quantities, never as a tracked reference in their own right.

5.2 The decision variable is the robot command, not the generator's parameters

The optimization variable is the robot command sequence U_k . The generator parameters θ are fixed during each experiment. Switching from impedance to admittance changes C_θ, G_θ , not the QP constraints or the command variable — stated formally as Theorem 1 (Section 6).

6. Basic Properties

Section 1 asserted, and Section 5.2 restated, that swapping the generator changes nothing in the QP except (C_θ, G_θ) . This section states that claim as a theorem and proves it, then establishes what the resulting QP guarantees in principle — convexity, exact realization when unconstrained, one-step constraint satisfaction — and, Proposition 3 in particular, what it does not.

Theorem 1 (Affine Generator Template Invariance). Fix the plant (A, B, m_r) , horizon N , weights $(W, \lambda_u, \lambda_{\Delta u})$, and bounds $(u_{\max}, \dot{u}_{\max}, p_{\max}, v_{\max})$. There is a single map $(C, G) \mapsto Q(C, G)$ — a quadratic-program template depending only on the plant, horizon, weights, and bounds above, never on θ — such that, for every generator θ with affine law (C_θ, G_θ) , the realization QP of Section 5 is exactly $Q(C_\theta, G_\theta)$. Consequently, for any two generators θ_1, θ_2 , the instances $Q(C_{\theta_1}, G_{\theta_1})$ and $Q(C_{\theta_2}, G_{\theta_2})$ share the same decision variable U_k and the same feasible set, and differ only in the numerical objective coefficients contributed by (C_θ, G_θ) , not in the decision-variable dimensions, constraint set, or objective template. Realizing a different affine generator therefore changes only the supplied (C_θ, G_θ) . This is stated for the affine generator class of Definition 1; a nonlinear generator (Section 4.3) makes $a_{i|k}^{\text{id}}$ a nonlinear function of U_k , so $\|r_{i|k}\|_W^2$ is no longer guaranteed convex quadratic and the template Q does not apply without per-step linearization.

Proof. Every constraint in Section 5's QP — the dynamics recursion, the acceleration definition, and the four box constraints — is written in terms of $A, B, m_r, u_{\max}, \dot{u}_{\max}, p_{\max}, v_{\max}, U_k$, and the resulting state/acceleration sequence; none references θ, C_θ , or G_θ . The feasible set is therefore a fixed polyhedron, identical for every generator. The only appearance of (C_θ, G_θ) anywhere in the problem is the term $a_{i|k}^{\text{id}} = C_\theta x_{i|k} + G_\theta \hat{F}_{h,i|k}$ inside the primary cost, entering exactly once per horizon step as an affine substitution into $r_{i|k} = a_{i|k} - a_{i|k}^{\text{id}}$. Because $x_{i|k}$ is itself independent of (C_θ, G_θ) and affine in U_k (established in Proposition 1's proof below), $a_{i|k}^{\text{id}}$ is affine in U_k with (C_θ, G_θ) -dependent coefficients only, so $\|r_{i|k}\|_W^2$ is a convex quadratic in U_k whose Hessian and gradient contributions are determined entirely by (C_θ, G_θ) ; the secondary terms $\lambda_u \|u_{i|k}\|_2^2$ and $\lambda_{\Delta u} \|u_{i|k} - u_{i-1|k}\|_2^2$ do not involve (C_θ, G_θ) at all. Collecting these pieces defines $Q(C, G)$ as a function of (C, G) alone — the plant, horizon, weights, and bounds enter Q as fixed parameters, not arguments — and by construction $Q(C_\theta, G_\theta)$ is exactly the QP Section 5 poses for generator θ . ■

This is why Section 4's two validated rows are checkable rather than aspirational: each supplies a different (C_θ, G_θ) into the one template Q above, and the theorem is precisely the statement that nothing else needs to change. The three sketched extensions of Section 4.3 are future work for exactly the reason the theorem's affine hypothesis makes explicit: without per-step linearization, f_θ does not factor through a fixed (C, G) pair, and Q does not apply.

Proposition 1 (convexity for affine generators). If $W \succeq 0$, $\lambda_u \geq 0$, and $\lambda_{\Delta u} \geq 0$, then the finite-horizon problem in Section 5 is a convex quadratic program. It is strictly convex if the assembled Hessian is positive definite, including the common case $\lambda_u > 0$.

Proof. The lifted state $x_{i|k}$ is affine in U_k by the recursion $x_{i+1|k} = Ax_{i|k} + B(u_{i|k} + \hat{F}_{h,i|k})$ unrolled from $x_{0|k}$. Hence $a_{i|k} = (u_{i|k} + \hat{F}_{h,i|k})/m_r$ and $a_{i|k}^{\text{id}} = C_\theta x_{i|k} + G_\theta \hat{F}_{h,i|k}$ are affine in U_k , so $r_{i|k} = a_{i|k} - a_{i|k}^{\text{id}}$ is affine in U_k and $\|r_{i|k}\|_W^2$ is convex quadratic since $W \succeq 0$. The regularization terms $\lambda_u \|u_{i|k}\|_2^2$ and $\lambda_{\Delta u} \|u_{i|k} - u_{i-1|k}\|_2^2$ are convex quadratic in U_k for $\lambda_u, \lambda_{\Delta u} \geq 0$, and a sum of convex functions is convex. Every listed constraint bounds an affine function of U_k in absolute value, so the feasible set is an intersection of half-spaces, hence a convex polyhedron. ■

Proposition 2 (exact unconstrained realization). Assume additionally that $W \succ 0$ (strictly positive definite — a stronger hypothesis than Proposition 1's $W \succeq 0$). Suppose an input sequence exists for which $r_{i|k} = 0$ over the horizon and no constraint is active. If the realization residual is optimized lexicographically before secondary effort objectives—or equivalently the secondary weights are zero—then every optimal primary solution realizes the reference dynamics exactly.

Proof. The primary objective $\sum_i \|r_{i|k}\|_W^2$ is nonnegative and attains the value zero at the assumed feasible sequence, so its minimum over the horizon is zero. Under lexicographic priority (or zero secondary weights), any optimal solution must attain this minimum, so $\sum_i \|r_{i|k}\|_W^2 = 0$, and since

each term is nonnegative, $\|r_{i|k}\|_W^2 = 0$ for every i . Because $W \succ 0$, a quadratic form $\|r\|_W^2$ vanishes only at $r = 0$; if W were only positive semi-definite, this step would give $r_{i|k} \in \ker W$ rather than $r_{i|k} = 0$, which is why the stronger hypothesis is needed here but not in Proposition 1. ■

With finite scalar weights, effort regularization can intentionally introduce a small residual even when exact realization is feasible. The present experiment uses small, not zero, secondary weights, so the predictive controller's measured residual is not exactly zero even during unconstrained intervals (Table 1) — this is the finite-weight case Proposition 2 itself distinguishes from the idealized zero-weight one, not a violation of it. The reactive comparator applies the generator's instantaneous law directly, with no horizon and no secondary regularization to trade against; its residual sits at numerical-solver tolerance (Table 1) for the simpler reason that it is solving a different, unconstrained algebraic inversion, not the same QP with different weights. That near-exactness is a check that the underlying generator and dynamics model are implemented correctly, not a demonstration of Proposition 2 itself.

Proposition 3 (one-step constraint inheritance). Assume the model and current human force are exact over the executed sample and the QP is feasible. Then $u_{0|k}^*$, $p_{1|k}$, and $v_{1|k}$ satisfy their corresponding QP bounds. Re-solving at every sample yields constraint satisfaction by induction while feasibility is retained.

This proposition is not a recursive-feasibility theorem. A deployable version requires a terminal invariant set, soft constraints with a quantified fallback, or a backup safe controller. Sudden force changes inside one sample and model error must also be covered by robust constraint tightening.

7. Simulation

Section 6 establishes what the QP guarantees in principle — convexity, exact realization when unconstrained, one-step constraint satisfaction, and what it does not guarantee (recursive feasibility). This section tests those properties empirically on the planar plant of Section 3.2, comparing predictive realization against the reactive comparator under identical command and rate limits.

7.1 Reproducible setup

The simulation uses a 2.5 kg planar point mass, $\Delta t = 0.02$ s, and a 20-step (0.4 s) horizon. A smooth 12 N force is applied along $+y$ from 1 s to 3 s. Limits are

$$|u_j| \leq 18 \text{ N}, \quad |v_j| \leq 0.22 \text{ m/s}, \quad |p_j| \leq 0.10 \text{ m},$$

with $|\dot{u}_j| \leq 180 \text{ N/s}$. The realization, force, and force-rate weights are 1, 2×10^{-4} , and 10^{-3} , respectively.

The impedance generator (Section 4.1) uses

$$M_d = 2.0 \text{ kg}, \quad D_d = 18 \text{ Ns/m}, \quad K_d = 45 \text{ N/m}.$$

The admittance generator (Section 4.2) uses $T_a = 0.25 \text{ s}$ and $Y = 0.025 \text{ m/(Ns)}$.

The comparator computes the instantaneous reference acceleration, converts it to robot force, and applies the same 18 N force and 180 N/s slew limits. It has no prediction of position or speed constraints. This comparator is intentionally minimal: it isolates what horizon-wide state constraints change; it is not presented as a survey-complete baseline.

7.2 Results

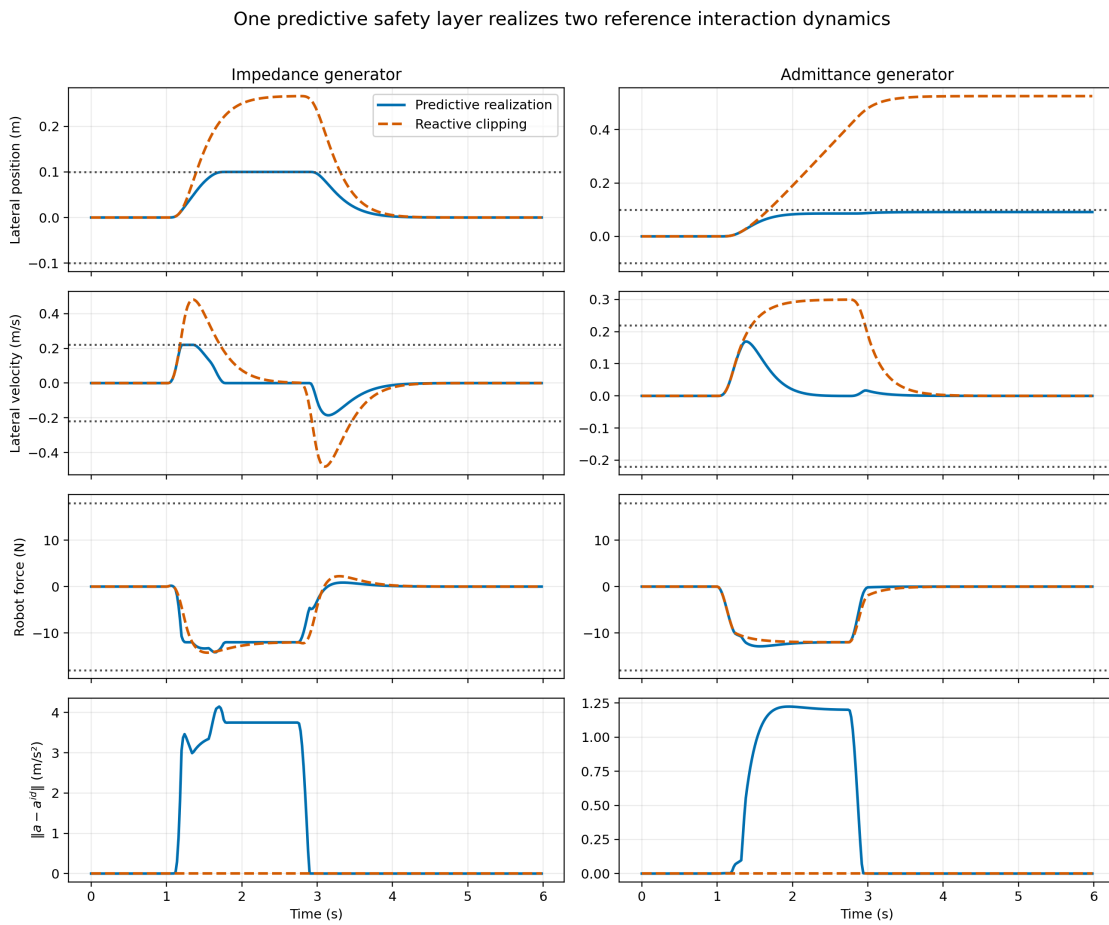


Figure 2. The same predictive realization layer is used with impedance (left) and force-guided admittance (right). Dotted lines are component-wise limits. Reactive clipping follows the requested behavioral acceleration almost exactly but does not anticipate state-limit violations. Predictive realization departs from the generator when necessary, visible in the final row.

Generator	Controller	Realization RMSE, componentwise (m/s ²)	Peak $ p_j $ (m)	Peak speed (m/s)	Peak $ u_j $ (N)	State-limit violation
Impedance	Predictive realization	1.351	0.100002	0.220	14.16	0.000002 m numerical tolerance
Impedance	Reactive clipping	$< 10^{-6}$	0.2663	0.480	14.24	0.1663 m, 0.260 m/s
Admittance	Predictive realization	0.405	0.0912	0.169	12.88	none
Admittance	Reactive clipping	$< 10^{-6}$	0.5250	0.300	12.00	0.4250 m, 0.080 m/s

Table 1. State-limit violation reports the amount by which the 0.10 m / 0.22 m/s workspace/speed bound is exceeded. Realization RMSE for the reactive comparator is at numerical-solver tolerance because it directly inverts the generator's instantaneous law for the force that realizes it exactly, with no horizon or secondary regularization to trade against; it is not zero for predictive realization because the horizon-wide formulation trades a small residual for constraint satisfaction and finite secondary weights (Proposition 2, Section 6), and because the reported metric aggregates over the whole run, not only the constrained interval. This RMSE is *componentwise*: the residual's two components and all time samples are pooled into a single mean before the square root. Table 2's FR3 residual is instead a *vector-norm* RMSE (each sample's 3-vector residual is norm-squared before averaging over time); the two are different quantities and are not directly comparable in magnitude across the two studies.

The impedance reference has a static displacement $F_h/K_d = 12/45 = 0.267$ m, matching the 0.266 m simulated peak after the smooth force ramp. Because this behavior is incompatible with the 0.10 m workspace, the predictive controller increases the realization residual while the bound is active. After release, it returns to the impedance equilibrium at the origin.

The admittance reference requests approximately 0.30 m/s under 12 N and has no restoring spring. Reactive realization therefore accumulates 0.525 m displacement and retains it after release. The constrained controller begins departing from the requested velocity dynamics before reaching the

workspace boundary and settles at 0.091 m. The generator remains unchanged; the behavior modification is entirely attributable to the realization layer.

7.3 Online generator switching

Sections 7.1–7.2 validate the generator interface by running two separate simulations, one per generator — informative, but not yet a demonstration that a single running controller accepts a different generator without redesign. This section tests that directly. One `InteractionDynamicsMPC` instance is constructed once, with the impedance generator; a small constant lateral force (1 N, well inside every bound) is applied for the entire 6 s run, with no release. At $t = 2$ s and $t = 4$ s, exactly one attribute is reassigned on the live controller — `controller.generator` — switching impedance $\rightarrow$ admittance $\rightarrow$ impedance; nothing else (the QP weights, constraints, decision variable, or the `previous_command` state carried between solves) is touched or reconstructed.

Online generator switching: one controller instance, only `.generator` reassigned

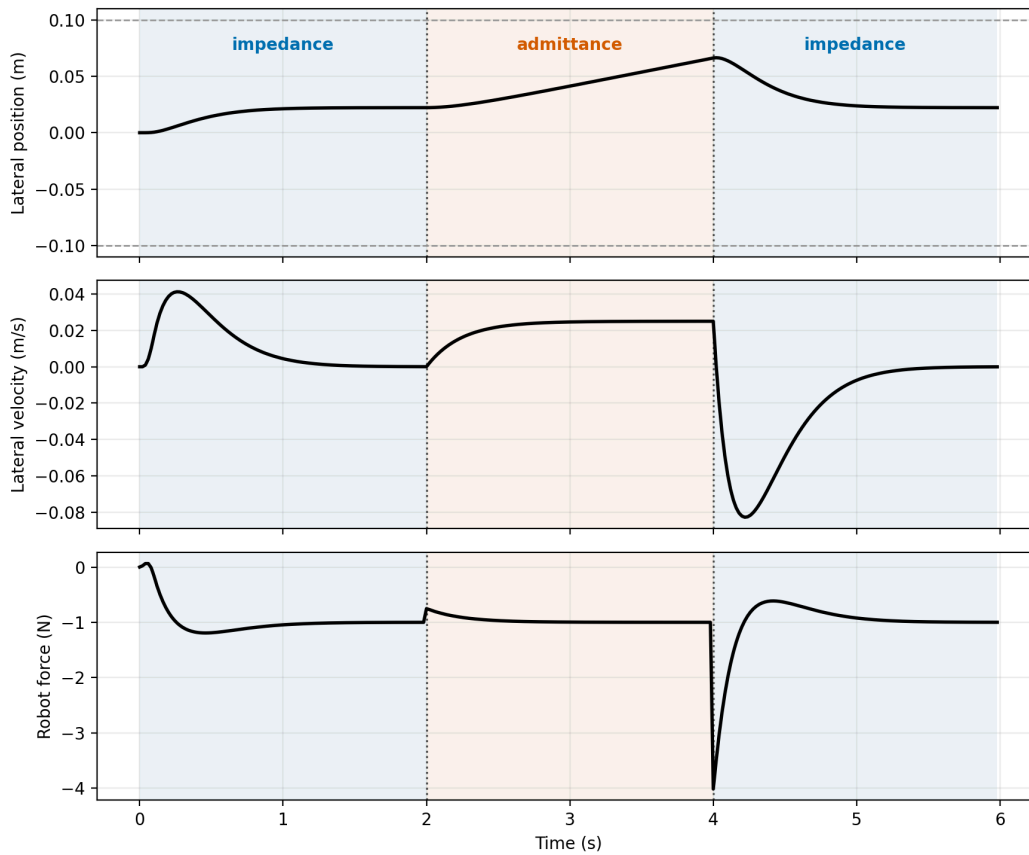


Figure 3. One controller instance; only `.generator` is reassigned at $t = 2$ s and $t = 4$ s (dotted lines). Position converges to the impedance equilibrium $F_h/K_d \approx 0.022$ m during both impedance segments, drifts at the admittance generator's steady-state velocity $Y F_h = 0.025$ m/s with no restoring term during the middle segment, and re-converges to the same equilibrium after re-entering impedance from a different state (nonzero position and velocity) at $t = 4$ s.

Both switches are rate-bounded: the largest command-force jump at a switch instant is 3.02 N at $t = 4$ s, 84% of the QP's 3.6 N per-tick limit. This is a material command change, not evidence of mathematical smoothness; it simply satisfies the same discrete rate constraint that bounds every other step. Position and speed stay within 0.067 m and 0.083 m of the origin throughout, well inside the 0.10 m / 0.22 m/s bounds. The experiment therefore validates the software-level generator interface and feasibility of an online swap under the existing rate constraint; it does not establish optimal switching, hybrid stability, or continuity of the generator law. The second impedance segment additionally shows re-convergence from the different position and nonzero velocity left by the admittance segment without special-casing on re-entry.

7.4 What the experiments establish

The results support four limited claims:

1. two qualitatively different affine interaction models use the same constrained predictive implementation;
2. an explicit realization residual identifies when safety requires deviation from the requested behavioral acceleration;
3. horizon-wide state constraints prevent violations that command clipping alone cannot prevent in this deterministic model;
4. a single controller instance accepts a different generator online, mid-run, by reassigning one attribute, with its command change bounded by the existing rate constraint.

They do not establish manipulator-level feasibility, coupled human–robot stability, passivity, robustness to force-estimation delay, or real-time performance on embedded hardware.

8. FR3 Manipulator Validation

Section 7 isolates the generator–realizer distinction from kinematic and inverse-dynamics confounds by using a point mass. This section instantiates the same architecture on a torque-controlled 7-DOF Franka FR3 simulated in MuJoCo, replacing the exact double-integrator plant with the real, configuration-dependent manipulator dynamics of Section 3.1.

8.1 Architecture

At each solve, let $J_v(q) \in \mathbb{R}^{3 \times 7}$ be the translational Jacobian and $\Lambda(q)^{-1} = J_v M(q)^{-1} J_v^\top$ the (regularized) inverse operational-space inertia for the position coordinate.

Full closed-loop law. Let R_d be the held reference orientation, $e_R(q) \in \mathbb{R}^3$ the corresponding orientation error, $J_w(q) \in \mathbb{R}^{3 \times 7}$ the rotational Jacobian, and q_0 a fixed neutral configuration. The commanded joint torque is assembled from four terms:

$$\begin{aligned}
\tau_{\text{ff}}(q, \dot{q}) &= C(q, \dot{q})\dot{q} + g(q) && \text{(feedforward)} \\
\tau_{\text{orient}}(q, \dot{q}) &= J_w(q)^\top (-K_{\text{rot}} e_R(q) - D_{\text{rot}} \omega) && \text{(orientation hold, raw)} \\
\tau_{\text{null}}(q, \dot{q}) &= -k_{\text{null}}(q - q_0) - d_{\text{null}} \dot{q} && \text{(null-space centering, raw)} \\
\bar{N}_v(q) &= I_7 - M(q)^{-1} J_v(q)^\top \Lambda(q) J_v(q) && \text{(position-consistent null-space projector)} \\
\tau_{\text{aux}}(q, \dot{q}) &= \bar{N}_v(q)^\top [\tau_{\text{orient}}(q, \dot{q}) + \tau_{\text{null}}(q, \dot{q})] && \text{(projected, see below)} \\
\tau_{\text{base}}(q, \dot{q}) &= \tau_{\text{ff}}(q, \dot{q}) + \tau_{\text{aux}}(q, \dot{q})
\end{aligned}$$

and the executed torque is

$$\tau = \tau_{\text{base}}(q, \dot{q}) + J_v(q)^\top F_{\text{cmd}}.$$

Task acceleration obeys $\ddot{y} = J_v \ddot{q} + \dot{J}_v \dot{q}$. Consequently, define

$$d_{\text{known}}(q, \dot{q}) = J_v(q) M(q)^{-1} \tau_{\text{aux}}(q, \dot{q}) + \dot{J}_v(q, \dot{q}) \dot{q},$$

which contains both the projected auxiliary-torque contribution and the task-kinematic term. The residual translational dynamics used by the QP are

$$\ddot{e} = \Lambda(q)^{-1} (F_{\text{cmd}} + F_h) + d_{\text{known}},$$

which corrects the sign of the sketch in the earlier draft of this section: the realized acceleration is *positive* in the commanded force, matching the point-mass convention of Section 3.2. $F_{\text{cmd}} \in \mathbb{R}^3$, the QP's Cartesian correction force, is supplied by whichever controller is active. The predictive controller sets $F_{\text{cmd}} = F_{0|k}^*$, the first element of the receding-horizon solution (Section 5, condensed later in this section). The reactive comparator instead sets

$$F_{\text{cmd}} = \text{clip} \left(\Lambda(q) [a^{\text{id}}(x, F_h) - d_{\text{known}}] - F_h \right),$$

rate- and magnitude-limited to the same bounds as the predictive controller, with $a^{\text{id}} = C_\theta x + G_\theta F_h$ the generator's instantaneous behavioral acceleration (Section 4). Both controllers share the identical τ_{base} and the identical torque-assembly step; they differ only in how F_{cmd} is chosen.

$\bar{N}_v(q)$ above is not the only choice that projects the auxiliary torque away from the translation task, and getting this projection right mattered in practice. A first implementation projected the null-space torque through the null-space projector of the *full* 6-row Jacobian and left the orientation-hold torque unprojected. Both choices leaked into $\ddot{e}$. Under sustained interaction this produced a positive-feedback path: joint drift increased the auxiliary torque, which increased translational leakage and the

compensating command. Applying the dynamically consistent position-task projector to both auxiliary terms removes that structural path [9]. For an exact inverse, $J_v M^{-1} \bar{N}_v^\top = 0$. The implementation uses Tikhonov regularization when forming Λ , however, so orthogonality is approximate rather than exact; the remaining projected-torque leakage is retained explicitly in d_{known} , together with $\dot{J}_v \dot{q}$.

This remaining leakage is small at any single tick but one-directional. A weak null-space centering spring lets it accumulate: over a full multi-second run it can drive the joint configuration more than a radian from q_0 , which in turn grows d_{known} further and inflates the correction the controller must apply — the same feedback structure as the leakage path above, just slow enough to be invisible on a short test and to show up only over the full benchmark duration.

This is confirmed by a saved gain sweep (`simulation/sweep_null_space_gains.py` , `results/null_space_gain_sweep.json`) over all four benchmark conditions, tracking configuration deviation and torque margin over the full 6 s run rather than an early window. At the original $k_{\text{null}} = 10$, $d_{\text{null}} = 2$, deviation exceeds 1.3 rad in every condition (1.32–1.85 rad), and the reactive (clipped) impedance condition — the one path with no QP and hence no hard torque enforcement — is actuator-infeasible, exceeding a per-joint torque limit by about 1.6 Nm; the predictive conditions never violate at any gain in this sweep.

$k_{\text{null}} = 40$, $d_{\text{null}} = 8$ is the gentlest increase in the sweep that clears the torque limit and keeps deviation under 0.29 rad in every condition, but it is not damping-free. Relative to the 10/2 baseline it substantially changes both reactive generators' own task-space response, through the same approximate, regularization-limited coupling responsible for the drift itself: admittance's reactive peak falls from 0.474 m to 0.107 m, and impedance's reactive peak falls from 0.158 m to 0.104 m. The predictive conditions barely move (0.061 m to 0.060 m and 0.062 m to 0.060 m), because their peak displacement is set by the workspace bound rather than the null-space term.

Stronger gains still ($k_{\text{null}} = 100$, $d_{\text{null}} = 20$) reduce deviation further (under 0.19 rad) and continue to suppress the admittance generator's task-space response specifically: its predictive peak drops to 0.044 m, now visibly below the 0.06 m bound rather than sitting against it, and its reactive peak falls to 0.058 m. The impedance conditions are comparatively insensitive beyond 40/8 (reactive peak 0.104 m to 0.102 m). 40/8 is used because it is the gentlest gain that clears the torque limit and controls drift in every condition; it trades a smaller, but real, task-space effect for that margin, and the sweep does not support a claim that any tested gain leaves the generators' task-space response undisturbed.

The realization QP freezes $\Lambda(q)^{-1}$, $d_{\text{known}}(q, \dot{q})$, $\tau_{\text{base}}(q, \dot{q})$, and $J_v(q)$ at the current solve and holds them across the horizon. In compact form, its decision vector contains the Cartesian command sequence and nonnegative state slacks,

$$\begin{aligned}
\min_{\{F_i, s_i^p, s_i^v\}_{i=0}^{N-1}} \quad & \sum_{i=0}^{N-1} \left(\|\ddot{e}_i - a_i^{\text{id}}\|_W^2 + \lambda_F \|F_i\|_2^2 + \lambda_{\Delta F} \|F_i - F_{i-1}\|_2^2 + \rho((s_i^p)^2 + (s_i^v)^2) \right), \\
\ddot{e}_i = \quad & \Lambda_k^{-1}(F_i + \hat{F}_{h,i}) + d_{\text{known},k}, \\
|F_i| \leq \quad & F_{\text{max}}, \quad |F_i - F_{i-1}| \leq \Delta F_{\text{max}}, \\
|\tau_{\text{base},k} + J_{v,k}^\top F_i| \leq \quad & \tau_{\text{max}}, \\
|e_i| \leq e_{\text{max}} + s_i^p, \quad |\dot{e}_i| \leq v_{\text{max}} + s_i^v, \quad & s_i^p, s_i^v \geq 0.
\end{aligned}$$

This per-solve model is a convex QP but only a local predictor of the nonlinear MuJoCo plant. The hard torque constraint is imposed at every predicted step, rather than only $i = 0$. It guarantees feasibility of the frozen-model command sequence returned by a successful solve; it is not a horizon-wide guarantee for the future nonlinear trajectory. Executed torque is therefore recomputed and monitored at 1 kHz.

This bound is on the *sum* $\tau_{\text{base}} + J_v^\top F_{\text{cmd},i}$, not on $F_{\text{cmd},i}$ alone: τ_{base} can itself consume a substantial share of τ_{max} in an unfavorable configuration (near a joint limit, or during an orientation-hold transient), leaving less budget for the interaction correction than a bound on $F_{\text{cmd},i}$ alone would assume. This is the actuator-saturation risk motivated in Section 1, applied here: the bound is part of the QP's feasible set rather than a check on its output, so any sequence the solver returns satisfies it by construction — the mechanistic contrast with a PD or impedance law's after-the-fact clipping is developed there and not repeated here. When the generator's requested behavioral acceleration would require more torque than the budget allows, the optimizer instead returns the closest feasible F_{cmd} , and that trade-off is exactly the realization residual r_k of Section 3.4: reported, not hidden.

The Cartesian workspace and speed box is slack-relaxed with a large quadratic penalty ($\rho = 10^8$). Under the frozen-Jacobian approximation, a hard box has no recursive-feasibility guarantee for the true nonlinear dynamics. The slack penalty keeps the optimization defined when the local prediction cannot satisfy the box exactly while making violation costly. The slack-relaxed constraint is genuinely exercised in the reported benchmark, not merely present in the formulation: 110 of the impedance-predictive solves and 203 of the admittance-predictive solves report nontrivial slack, with a peak position slack of about 0.05 mm and a peak speed slack below 1×10^{-6} m/s in both conditions -- small in magnitude, but nonzero, and now logged and reported directly (`run_fr3_experiments.py`) rather than only inferred from the position bound being held. Separately, if a solve is reported infeasible, the implementation falls back to the rate- and magnitude-limited reactive law; that fallback path was not triggered in the reported benchmark (zero infeasible solves in every condition), so it is exercised only by the slack mechanism above, not by the reactive fallback itself.

The fast control loop recomputes τ_{base} , the orientation term, and the null-space term from the current $(q, \dot{q})$ at 1 kHz; the QP is re-solved only every 20 ms (a 50 Hz outer loop), holding F_{cmd} between

solves. Caching the full joint torque at the outer-loop rate, rather than only the slow correction term, would confound any comparison of predictive against reactive control at a common inner-loop rate.

8.2 Benchmark scenario

Mirroring the planar prototype's own story rather than a separate benchmark design: the EE holds a fixed nominal pose and orientation while a sustained human push toward a workspace/speed boundary is applied, and predictive realization is compared against reactive clipping under an identical command/rate box. Neither controller has oracle knowledge of the scripted force; both receive a zero-order-hold forecast of the currently measured force, as in Section 7.

The push is a 20 N force along $-z$ with a raised-cosine ramp from $t = 1.0$ s to 1.25 s, a hold to $t = 3.25$ s, and a symmetric ramp-down to $t = 3.5$ s. It is applied physically to the MuJoCo end-effector body, not only supplied to the controller's internal model, so the comparison reflects what the real simulated dynamics do under the push, not only what each controller believes is happening. The 2 s hold (versus an earlier 0.5 s) gives the admittance generator's velocity time to build toward its steady state (time constant $T_a = 0.3$ s below); at the shorter hold, the corrected dynamics above show the reactive comparator does not approach the workspace bound at all, so no genuine predictive-versus-reactive contrast exists to report for that generator. The impedance generator's story is unaffected by the hold length, since its peak is set by its equilibrium, reached well within 0.5 s (confirmed directly at both hold values).

The workspace/speed bound is

$$|e_z| \leq 0.06 \text{ m}, \quad |v| \leq 0.35 \text{ m/s},$$

the QP horizon is 15 steps (0.3 s) at $\Delta t = 0.02$ s, and per-joint torque limits match the FR3 (± 87 Nm for joints 1–4, ± 12 Nm for joints 5–7).

The impedance generator (Section 4.1) uses $M_d = 2.0$ kg, $D_d = 28$ Ns/m, $K_d = 200$ N/m. The admittance generator (Section 4.2) uses $T_a = 0.3$ s and $Y = 0.01$ m/(Ns). Each condition runs for 6 s of simulated time.

8.3 Results

One predictive realization layer on a real FR3: two interaction dynamics, horizon-wide torque feasibility

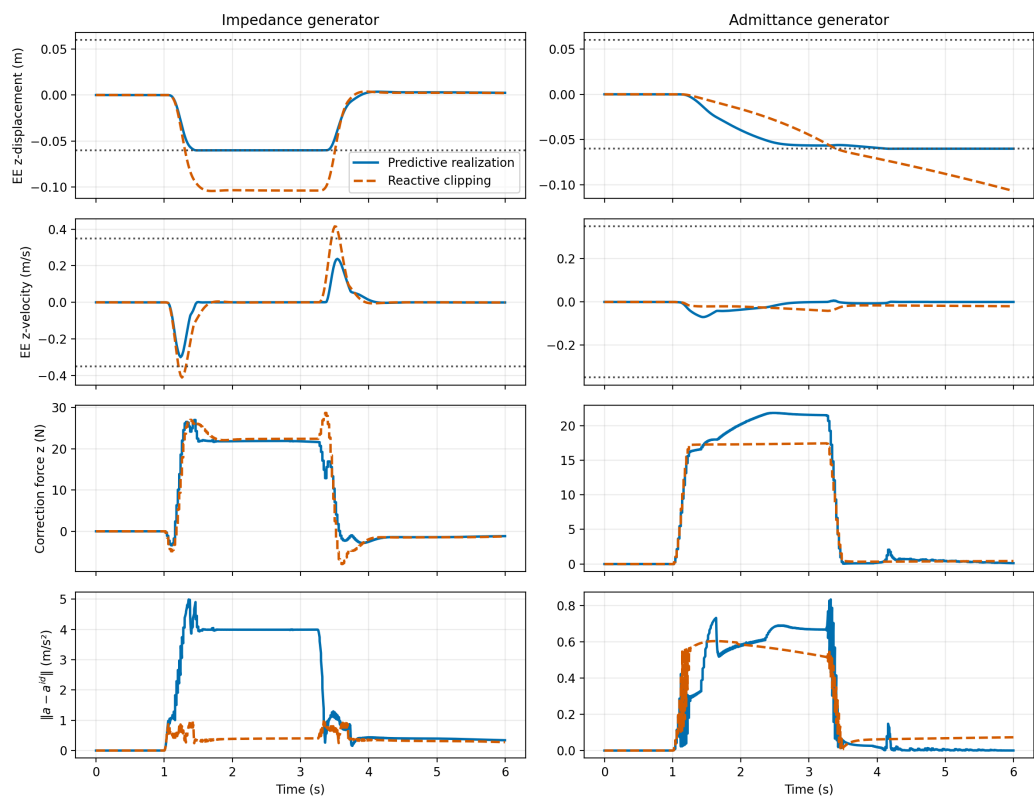


Figure 4. The same predictive realization layer on a real FR3, impedance (left) and admittance (right) generators, predictive realization (solid) against reactive clipping (dashed). Dotted lines are the workspace/speed bounds. Predictive realization tracks the bound closely for the duration of the push; reactive clipping overshoots substantially, and for the admittance generator does not recover the displacement after force release.

Generator	Controller	Empirical / predicted residual RMSE, vector- norm (m/s ²)	Peak $ e_z $ (m)	Peak component speed (m/s)	Maximum torque utilization	State- limit violation
Impedance	Predictive realization	2.41 / 2.50	0.0602	0.300	37.0%	0.0002 m
Impedance	Reactive clipping	0.387 / 0.107	0.1044	0.415	37.2%	0.0444 m, 0.065 m/s

Generator	Controller	Empirical / predicted residual RMSE, vector- norm (m/s ²)	Peak $ e_z $ (m)	Peak component speed (m/s)	Maximum torque utilization	State- limit violation
Admittance	Predictive realization	0.366 / 0.502	0.0601	0.070	34.4%	0.0001 m
Admittance	Reactive clipping	0.346 / 0.059	0.1066	0.041	31.7%	0.0466 m

Table 2. The *predicted* residual uses the frozen local model in Section 8.1. The *empirical* residual instead finite-differences the 1 kHz MuJoCo end-effector velocity and compares that acceleration with the generator request; it is the appropriate measure of behavior delivered by the simulated nonlinear plant. Their gap measures local-model and differentiation error, so the predicted residual must not be interpreted as an exact plant residual. Both are vector-norm RMSE (Table 1's caption), not the componentwise RMSE reported for the planar study; the two are not directly comparable in magnitude. Speed is reported component-wise to match the box constraint. Maximum torque utilization is the largest ratio $|\tau_j|/\tau_{\max,j}$, which avoids comparing unlike 87 Nm and 12 Nm joint limits. Predictive solve times are reported in the text below because the reactive comparator does not solve a QP.

No condition violates a per-joint torque limit at the executed sample, no solve is reported infeasible, and the torque constraint is never the one that actively binds in this benchmark: maximum torque utilization stays below 38% in every condition, well short of saturation. This benchmark does not include a first-step-only ablation, so it does not by itself isolate the performance benefit of enforcing that constraint over the full horizon rather than only at the first step. Correct horizon-wide enforcement is verified separately by a binding unit test

(`test_horizon_wide_torque_feasibility_binding` in `simulation/test_fr3_interaction_dynamics_mpc.py`) that artificially tightens one joint's limit until the constraint is forced to activate; that test confirms the mechanism works as specified, not that it outperforms a first-step-only formulation. The performance advantage of horizon-wide over first-step-only enforcement remains to be evaluated through a matched stress-case ablation.

The impedance reference has an unconstrained static displacement $F_h/K_d = 20/200 = 0.10$ m, which by itself already exceeds the 0.06 m bound. Reactive clipping tracks toward that equilibrium and peaks at 0.104 m (slightly above the static value, from the same ramp-overshoot effect as the

planar case in Section 7.2), overshooting the bound by 4.4 cm. Predictive realization instead departs from the requested behavioral acceleration while the bound is active, holding displacement to 0.0602 m.

The admittance reference has no equilibrium displacement to compare against, only a steady-state velocity $Y F_h = 0.01 \times 20 = 0.2$ m/s while the force is held, with no restoring term to pull it back after release. Reactive clipping accumulates displacement through the hold and continues briefly after force release before the velocity decays (time constant $T_a = 0.3$ s), peaking at 0.1066 m — a 4.7 cm overshoot — and never recovers it, since the generator itself has no position-restoring term (Section 4.2) and the reactive comparator has no lookahead to anticipate the boundary. Predictive realization begins departing from the requested velocity before the bound is reached and stays within 0.0601 m.

One limitation is visible in the results and worth reporting rather than hiding: the admittance-generator QP's solve time (mean ~11.6-11.8 ms, max ~106-110 ms across repeated runs on the development machine) is within the nominal 20 ms outer-loop period on average but exceeds it substantially at its worst tick; the impedance generator's solves (mean ~4.4 ms, max 7-17 ms across repeated runs) rarely approach it. Both use the same dimensions and constraint structure but different generator-dependent objective coefficients. Conditioning is a plausible explanation for the mean difference, but it is not isolated experimentally. Solve time is wall-clock and machine-dependent, not bit-for-bit reproducible like the other quantities in Table 2 (confirmed by rerunning the benchmark four times: means were stable to within about 5%, but the observed maximum varied by more than 2x run to run, since it is set by a single slow tick and is sensitive to OS scheduling noise). The loop is simulation-time paced, so these numbers indicate solver effort on this machine rather than a real-time guarantee, and the specific mean/max values here should not be treated as exactly reproducible on other hardware.

An earlier version of this benchmark, run before the null-space centering fix described in Section 8.1, showed a second issue: slack chattering over a late-time window as the admittance condition sat near the workspace bound. That was traced to the same slow configuration drift Section 8.1 now describes and controls; it is not present in the results above (Figure 4), and is not reported as a current limitation.

8.4 Scope and what remains

This is a focused core, not the full manipulator validation program. The generator set is impedance and admittance only, not the nonlinear/passive generator of Section 4.3; the scenario is a single sustained push, not a sweep over delay, sensor noise, or mass mismatch; there is no collision constraint, and the reactive clipped comparator is intentionally weak. A publication-level evaluation needs matched classical impedance/admittance, control-barrier or reference-governor, predictive-

variable-impedance, model-predictive-impedance, and model-predictive-interaction baselines. The online generator-switching demonstration (Section 7.3) is planar only; an FR3 counterpart remains future work.

Human-participant and hardware validation remain future work, as does closing the admittance solve-time gap identified in Section 8.3.

9. Limitations and Open Theory

The results above establish the architectural distinction on two plants; they do not establish everything a mature interaction-dynamics realization framework would need. Five gaps are worth stating plainly rather than leaving implicit.

First, the current generator class is affine and memoryless (Section 4.3 sketches richer instances, none yet implemented). A stateful or learned generator requires augmented prediction and a clear treatment of model uncertainty.

Second, the realization cost does not itself imply passivity. A passive reference generator can still be rendered non-passive by constraint-induced deviation or secondary optimization. A useful extension is a dissipativity constraint on the realized port variables together with an energy tank or passivity observer/controller.

Third, hard state constraints can become infeasible after an unpredicted impulse — Proposition 3 already flags this as a gap, not a guarantee. The FR3 study (Section 8) addresses it operationally with a slack-relaxed workspace/speed box and a reactive fallback on solver infeasibility. Of these two mechanisms, only the slack relaxation was actually exercised in the tested scenario (Section 8.1: nontrivial slack on a substantial minority of predictive solves, always small in magnitude); the solver-infeasibility fallback was never triggered, since no solve was reported infeasible in this scenario. So the demonstration supports "the slack mechanism did not produce unbounded violations here," not "the fallback path has been exercised" — and even the slack claim is an empirical demonstration for one scenario, not the terminal invariant set, quantified soft-constraint fallback, or backup safe controller Proposition 3 says a deployable version would need. Softening safety-critical constraints without defining the resulting risk remains not sufficient on its own.

Fourth, the realization residual has physical units and depends on the chosen output coordinates, and the FR3 study only exercises it on the translational coordinate. Orientation on the FR3 is held by a fixed PD law, τ_{orient} (Section 8.1), not by a generator. The architectural claim is therefore validated for translation only. Extending the interface to orientation requires a principled metric or energy-weighted normalization across coordinates with different units. Even within one coordinate, raw RMSE values

across generators with different characteristic scales are not direct performance rankings. In addition, Table 2's empirical and frozen-model residuals differ materially in several conditions. That discrepancy is evidence that model adequacy and acceleration estimation must be reported separately, not a solver residual or a violation of Theorem 1.

Finally, the architectural novelty must be assessed against the full model predictive impedance/interaction-control literature surveyed in Section 2, not only the two closest precedents named there. The strongest eventual claim should rest on implemented generator modularity, formal residual guarantees, and matched experiments—not terminology alone.

10. Conclusion

This paper's central result is architectural: for memoryless affine generators, Theorem 1 establishes invariance of the QP template, decision-variable dimensions, and feasible set while (C_θ, G_θ) changes the numerical residual-cost coefficients. The simulations instantiate that claim for impedance and admittance. The planar study shows anticipatory state-constraint handling and an online generator swap whose command change respects the existing rate bound. The FR3 study shows the same interface on a nonlinear manipulator with frozen-model horizon constraints; it enforces joint torque feasibility at every horizon step and limits workspace overshoot relative to a reactive clipped comparator, though the torque bound is never the one that actively binds in this particular benchmark (Section 8.3). It also exposes an important qualification: the residual predicted by the local operational-space model can differ substantially from the residual measured from MuJoCo motion. The present evidence is therefore a reproducible simulation proof of concept, not a claim of universal generator compatibility, passivity, hardware safety, or superiority to established predictive interaction controllers. The next steps are a genuinely nonlinear or learned generator, stronger matched baselines, model-mismatch and delay sweeps, a first-step-versus-horizon torque-constraint ablation, and hardware or participant validation.

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Reproducibility

Run from `pHRI/imp_reference` :

```
# Architecture schematic (Figure 1)
python3 simulation/make_architecture_diagram.py

# Planar point-mass study (Section 7, Figure 2)
python3 simulation/run_experiments.py
python3 -m pytest simulation/test_interaction_dynamics_mpc.py simulation/test_benchmark_

# Online generator-switching demonstration (Section 7.3, Figure 3)
python3 simulation/run_generator_switching_experiment.py
python3 -m pytest simulation/test_generator_switching.py -q

# FR3/MuJoCo manipulator study (Section 8, Figure 4)
python3 simulation/run_fr3_experiments.py
python3 -m pytest simulation/test_fr3_interaction_dynamics_mpc.py simulation/test_fr3_b

# Null-space centering gain sweep backing the drift claim in Section 8.1
python3 simulation/sweep_null_space_gains.py
```

Both driver scripts use a fixed deterministic scenario and record every numeric parameter and reported metric in `results/metrics.json` and `results/fr3_metrics.json`, respectively, but neither asserts anything itself. The `test_interaction_dynamics_mpc.py` / `test_fr3_interaction_dynamics_mpc.py` suites check the QP's mechanics on individual, hand-picked states; they do not run the 6-second scripted-force scenario. Tables 1 and 2 report numbers from. `test_benchmark_verification.py` / `test_fr3_benchmark_verification.py` close that gap: they run the actual benchmark scenario end-to-end and assert its headline claims directly — no torque-limit violation in any condition, no unhandled solver infeasibility, predictive realization within a controlled envelope of the workspace bound, and reactive clipping overshooting it by a checked margin — so a future change that silently breaks a claim in this paper fails a test rather than only showing up as a quieter number in a table nobody re-checks. Solve-time claims are bounded generously rather than asserted exactly, since they are wall-clock and were confirmed empirically to vary run-to-run (the position/torque numbers, by contrast, are exactly reproducible: re-running both benchmarks produced bit-for-bit identical values before this paper's figures and tables were finalized). The FR3 study imports the shared MuJoCo/operational-space infrastructure from `pHRI/simulation` (`fr3_mujoco.py`, `fr3_impedance.py`, `so3_utils.py`) rather than duplicating it. `sweep_null_space_gains.py` produces `results/null_space_gain_sweep.json`, the saved artifact behind Section 8.1's specific numeric claims about configuration drift at each tested $(k_{\text{null}}, d_{\text{null}})$ pair; those claims were previously backed only by interactive sessions with no saved script or output.